

Modeling Life Insurance Lapse Timing with Machine Learning and Survival Analysis

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Abstract

This study investigates the use of modern survival modeling techniques to predict lapse timing within a large health insurance portfolio. We compare three approaches—Cox Proportional Hazards, Random Survival Forests, and the Cox-Time neural survival model—to evaluate how structural assumptions, nonlinear interactions, and time-dependent covariate effects influence predictive performance. Using concordance index, Brier Score, calibration analysis, and SHAP-based interpretability, we find that Cox-Time provides the strongest overall performance, offering both accurate survival probability estimates and flexible hazard modeling across policy durations. Although the empirical analysis is grounded in health insurance, the modeling framework is transferable to other insurance lines, with feature engineering adapted to product-specific characteristics. The results demonstrate that incorporating flexible, time-aware survival models can substantially improve lapse prediction and provide deeper insight into policyholder behavior over time.

Keywords

Survival Analysis, Machine Learning, Lapse Modeling, Insurance Analytics, Health Insurance, Predictive Modeling

1 Introduction

Lapse modeling continues to play a central role in actuarial work because it directly affects profitability, pricing assumptions, and long-term portfolio management. As insurers rely more heavily on data-driven decision-making, understanding not only *why* policies lapse but also *when* they lapse has become increasingly important. A lapse disrupts expected cash flows, introduces uncertainty into reserving and forecasting, and complicates the assumptions used in product design.

Traditional actuarial approaches typically rely on aggregated experience studies or parametric models such as the Cox Proportional Hazards model [10]. These methods are transparent and easy to interpret, but they assume relatively simple, linear relationships and proportional hazards. In practice, lapse behavior is shaped by a combination of demographic factors, policy characteristics, claim activity, and behavioral signals that interact in nonlinear ways. When several moving parts influence lapse timing simultaneously, classical techniques may struggle to capture the full structure of the data.

Machine learning offers a way to address this complexity. Modern survival models—particularly Random Survival Forests [11] and neural extensions of Cox models [12]—can learn nonlinear interactions, accommodate high-dimensional feature spaces, and

model heterogeneous risk patterns more effectively than traditional methods. However, actuarial applications require not only accuracy but also stability, interpretability, and robustness under different data conditions.

This study evaluates three survival modeling approaches on a real health insurance dataset containing policyholder demographics, claim histories, and engineered behavioral features: the Cox Proportional Hazards model, Random Survival Forests, and a neural time-dependent Cox-Time model. In addition to comparing predictive performance, the analysis incorporates robustness checks across multiple random seeds, sensitivity tests using reduced subsets of the data, and feature ablation experiments focusing on the dominant covariates. These methods provide insight into how each model behaves, how stable its predictions are, and which features contribute most to lapse risk. The goal is to identify modeling approaches that balance accuracy, robustness, and interpretability for practical use in actuarial and analytics workflows.

2 Literature Review

Early research on lapse prediction in insurance relied heavily on binary classification models. These approaches treated lapse as a static event rather than a time-dependent process, limiting their ability to capture when a lapse occurs and how risk evolves over time. Lessmann et al.[1] benchmarked 41 classification algorithms for credit scoring and demonstrated that modern ensemble methods consistently outperform logistic regression, highlighting the importance of model selection in customer-behavior prediction. Similar findings appear in actuarial applications: Kgare[2] and Mogusu et al.[3] show that machine-learning classifiers outperform traditional statistical models when predicting lapse likelihood, identifying key predictors such as premium structure, demographics, and payment behavior. However, these studies do not incorporate censoring or event timing, both of which are essential for actuarial lapse modeling.

Recognizing these limitations, recent work has shifted toward survival analysis as a more appropriate framework for modeling lapse timing. Aleandri and Eletti[4] compare Accelerated Failure Time models with Random Survival Forests (RSF) using a large credit-protection insurance dataset and find that RSF provides superior predictive performance, particularly in capturing nonlinearities and interactions. Andrade and Valencia[5] evaluate Cox Proportional Hazards (CoxPH), RSF, and Conditional Inference Forests on a real-world life insurance portfolio, concluding that machine-learning-based survival models outperform CoxPH in both discrimination and calibration. These studies demonstrate the growing value of censoring-aware, time-to-event modeling in actuarial science.

Beyond classical survival models, deep learning has emerged as a powerful tool for time-to-event prediction. Zhang et al.[6] introduce survival neural networks designed for longitudinal studies, including Time-dependent Survival Neural Networks (TSNN) and Recurrent Survival Neural Networks (RSNN). These models leverage feed-forward and Long Short-Term Memories (LSTM) architectures to capture nonlinear relationships and repeated measurements over time. While effective in medical and engineering applications, they require longitudinal covariates that are not available in many insurance datasets, including the static policy-level dataset used in this study.

A closely related line of actuarial research explores longitudinal survival modeling using tree-based methods. Valla[7] proposes a framework based on left-truncated and right-censored (LTRC) survival trees, LTRC forests, and mixed-effect tree-based models to incorporate dynamic policyholder trajectories. These methods improve lapse prediction and customer lifetime value estimation but rely on detailed historical policy movements that are not present in the dataset used here.

Recent comparative studies in healthcare provide valuable insight into the performance of modern survival models. Wang et al.[8] conduct a comprehensive benchmark of CoxPH, penalized Cox models, RSF, Gradient Boosting survival models, AutoScore Survival, DeepSurv, CoxTime, and DeepHit using a large emergency dataset. Their results show that deep learning models such as DeepSurv and CoxTime achieve the highest discrimination, while RSF provides strong calibration. Importantly, their study evaluates models using both the concordance index (C-index) and integrated Brier score (IBS), demonstrating the importance of assessing both discrimination and calibration—an approach adopted in the present work.

Recent work in insurance analytics highlights the value of survival based machine-learning methods for modeling lapse timing and handling censored policy data [4, 5]. Neural survival models [6, 8] extend these capabilities, though many rely on longitudinal covariates that are not available in static health-insurance portfolios. However, modern survival approaches—including CoxTime—remain largely unexplored for lapse prediction using policy level demographic, financial, and behavioral variables. Robustness checks, majority-feature ablation, and calibration metrics such as the Integrated Brier Score are also rarely evaluated in actuarial lapse studies. These gaps motivate the present work, which compares CoxPH, RSF, and CoxTime using both discrimination and calibration metrics on a large health-insurance dataset.

3 Dataset

The analysis is based on a publicly available non-life health insurance portfolio containing **228,711 raw policy records** and **42 variables** describing demographic characteristics, policy attributes, socioeconomic indicators, financial behavior, and claim experience [9]. Each row corresponds to a unique insured policy observed over a one-year period, and each column represents a distinct covariate relevant to lapse behavior.

3.1 Portfolio Composition

The raw dataset exhibits substantial demographic and structural heterogeneity. Policyholders range in age from 0 to 99 years (mean

44.4), with a balanced gender distribution (54% female, 46% male). Policies are distributed across multiple channels, with the “I” channel accounting for 63% of all contracts. Policy tenure variables—*seniority_insured* and *seniority_policy*—range from 0 to 86 years, with medians of 9–10 years, reflecting a mature portfolio with long-standing insureds.

3.2 Socioeconomic Indicators

The dataset includes a rich set of regional socioeconomic variables (*C_GI*, *C_II*, *C_IE_P*, *C_IE_S*, *C_IE_T*, *C_GE_P*, *C_GE_S*, *C_GE_T*), each measured on a 1–100 scale. These variables capture municipal income, education, employment, and geographic segmentation. However, they exhibit 5–6% missingness and inconsistent categorical formats (e.g., *C_H* and *C_C* contain multiple string levels), indicating the need for cleaning and standardization. Their distributions are highly skewed, with some indicators concentrated in lower socioeconomic tiers and others in upper tiers.

3.3 Financial and Utilization Variables

Premiums range from €33 to €22,318 (mean €852), demonstrating wide variation in product pricing. Annual claims cost is strongly right-skewed, with a median of €186 and a maximum of €87,000, consistent with typical health insurance utilization patterns where many policies incur no claims and a small number generate high-cost events. The dataset also includes utilization indicators such as the number of medical services and reimbursement levels.

3.4 Policy Structure and Target Variables

The raw data contains both insured-level and policy-level lapse dates. Because most policies remain active, lapse dates exhibit 73–76% missingness, producing a high censoring rate. Exposure time and duration are derived from effective and lapse dates, with raw durations spanning a wide range across the portfolio.

3.5 Data Quality Considerations

Several data quality issues are present in the raw dataset:

- **High missingness** in socioeconomic indicators and lapse dates.
- **Strong skewness** in premium and claims distributions.
- **Inconsistent categorical formats** across socioeconomic and structural variables.
- **Potential data leakage** from seniority variables, which encode time-in-force information directly from raw policy dates.
- **Outliers** in financial variables, particularly premiums and claims.

These issues motivate the pre-processing pipeline described in the next section, which includes missing-value handling, feature removal, normalization, and construction of time-to-event variables. The raw dataset’s size and heterogeneity make it well suited for evaluating both classical and machine-learning survival models, while its censoring structure and nonlinear relationships justify the use of flexible approaches such as Random Survival Forests and neural Cox-Time models.

4 Methods

This section describes the full modeling pipeline used to predict lapse timing in a health insurance portfolio. The workflow consists of three major components: data preparation, feature engineering, and model development. All models were trained using a stratified train-validation-test split and evaluated using both the concordance index (C-index) and the integrated Brier score, providing complementary measures of discrimination and calibration.

4.1 Data Preparation

The raw dataset was first cleaned and standardized to ensure consistency across variables. Column names were normalized by stripping whitespace and replacing spaces with underscores, and duplicate records were removed. Several variables representing policyholder demographics, product attributes, and socioeconomic indicators were cast to categorical types.

Because lapse is a time-to-event outcome, survival variables were constructed directly from policy dates. A censoring date was derived from the reporting period, and the lapse date was replaced with the censoring date for policies that remained active. Duration was computed in days and converted to years, with negative values clipped to zero. An event indicator was created to denote whether a lapse occurred within the observation window.

Identifier fields (policy ID, insured ID) and raw socioeconomic variables were removed. Records missing socioeconomic tier information were excluded to maintain consistency in engineered features.

To support model training, the dataset was split into training (60%), validation (20%), and test (20%) sets using stratification on the event indicator to preserve the lapse rate across splits. Continuous variables were standardized using a `StandardScaler`, including seniority measures, claim cost, centered premium, and insured population counts.

Multicollinearity was assessed using the Variance Inflation Factor (VIF). Although VIF values were not used to drop variables directly, this diagnostic informed later steps in the Cox model preparation. For the Cox model specifically, constant columns and duplicated columns were removed to prevent perfect multicollinearity and ensure stable coefficient estimation.

4.2 Feature Engineering

A comprehensive feature engineering pipeline was applied to capture behavioral, financial, and structural patterns relevant to lapse timing. The main engineered feature groups include:

- **Age Binning:** Policyholder age was grouped into seven behavioral age bands (e.g., 0–18, 19–25, 26–34), followed by one-hot encoding.
- **Premium Features:** Premium values were log-transformed to reduce skew, centered around the portfolio mean, and segmented into deciles using quantile cuts. All deciles were one-hot encoded.
- **Socioeconomic Tiers:** Eight socioeconomic indicators were converted to numeric values, imputed using medians, and segmented into five quantile-based tiers. Each tier was one-hot encoded.
- **Utilization Buckets:** Medical service utilization was grouped into buckets (U0–U4), separating zero-utilization policies from quartile-based nonzero utilization. These buckets were one-hot encoded.
- **Policy Structure Features:** Product design groups were one-hot encoded, and interaction terms were created between product design indicators and centered premium to capture nonlinear relationships between pricing and product structure.
- **Distribution Channel:** Distribution channel categories were one-hot encoded.
- **Reference Category Removal:** To avoid multicollinearity in the Cox model, one dummy variable from each one-hot encoded group was removed, establishing a baseline category.
- **Final Cleaning:** Raw date fields, original categorical variables, and intermediate engineered variables were dropped after feature construction. Boolean fields were converted to integers, and categorical fields such as gender, reimbursement, and new business were mapped to binary indicators.
- **Leakage Feature Removal:** Seniority variables (*seniority_insured*, *seniority_policy*) were excluded because they encode time-in-force directly from raw policy dates, creating target leakage and artificially inflating model performance.

4.3 Modeling Framework

Three survival models were evaluated: a penalized Cox Proportional Hazards model, a Random Survival Forest, and a neural Cox-Time model.

Cox Proportional Hazards Model

The Cox Proportional Hazards model represents the hazard for an individual with covariates $\mathbf{x}$ in stratum s at time t as

$$h(t | \mathbf{x}, s) = h_{0,s}(t) \exp(\boldsymbol{\beta}^\top \mathbf{x}),$$

where $h_{0,s}(t)$ is the stratum-specific baseline hazard and $\boldsymbol{\beta}$ is the vector of regression coefficients.

Under the proportional hazards assumption, the hazard ratio between two individuals $\mathbf{x}_a$ and $\mathbf{x}_b$ in the same stratum is

$$\frac{h(t | \mathbf{x}_a, s)}{h(t | \mathbf{x}_b, s)} = \exp(\boldsymbol{\beta}^\top (\mathbf{x}_a - \mathbf{x}_b)),$$

which does not depend on time.

The stratified partial likelihood is

$$L(\boldsymbol{\beta}) = \prod_s \prod_{\substack{i: \delta_i=1 \\ s_i=s}} \frac{\exp(\boldsymbol{\beta}^\top \mathbf{x}_i)}{\sum_{j \in R_s(t_i)} \exp(\boldsymbol{\beta}^\top \mathbf{x}_j)},$$

where δ_i is the event indicator, s_i is the stratum of individual i , and $R_s(t_i)$ is the stratum-specific risk set.

An L_2 -penalized Cox model maximizes

$$\hat{\boldsymbol{\beta}} = \arg \max_{\boldsymbol{\beta}} \left[\ell(\boldsymbol{\beta}) - \frac{\lambda}{2} \|\boldsymbol{\beta}\|_2^2 \right],$$

where $\ell(\boldsymbol{\beta}) = \log L(\boldsymbol{\beta})$ and $\lambda = 1.0$.

Random Survival Forest

Random Survival Forests extend random forests to right-censored data. Splits are chosen using the log-rank statistic. For a candidate split producing daughter nodes A_L and A_R , the standardized log-rank statistic is

$$\text{LR}(A_L, A_R) = \frac{\sum_k \left(d_{k,L} - \frac{Y_{k,L}}{Y_k} d_k \right)}{\sqrt{\sum_k \frac{Y_{k,L}}{Y_k} \left(1 - \frac{Y_{k,L}}{Y_k} \right) \frac{Y_k - d_k}{Y_k - 1} d_k}},$$

where $d_{k,L}$ and $Y_{k,L}$ are the number of events and number at risk in A_L at time t_k , and d_k and Y_k are the corresponding totals.

Each terminal node uses the Nelson–Aalen estimator:

$$\hat{H}_b(t | \mathbf{x}) = \sum_{t_k \leq t} \frac{d_{bk}}{Y_{bk}},$$

and the ensemble cumulative hazard is

$$\hat{H}_{\text{RSF}}(t | \mathbf{x}) = \frac{1}{B} \sum_{b=1}^B \hat{H}_b(t | \mathbf{x}).$$

The tree-specific survival function is

$$\hat{S}_b(t | \mathbf{x}) = \prod_{t_k \leq t} \left(1 - \frac{d_{bk}}{Y_{bk}} \right),$$

and the ensemble survival estimate is

$$\hat{S}_{\text{RSF}}(t | \mathbf{x}) = \frac{1}{B} \sum_{b=1}^B \hat{S}_b(t | \mathbf{x}).$$

Cox-Time Neural Survival Model

The Cox-Time model allows covariate effects to vary with time. The hazard is

$$h(t | \mathbf{x}) = h_0(t) \exp(f_\theta(\mathbf{x}, t)),$$

where $f_\theta(\mathbf{x}, t)$ is a neural network.

The hazard ratio between individuals $\mathbf{x}_a$ and $\mathbf{x}_b$ is

$$\frac{h(t | \mathbf{x}_a)}{h(t | \mathbf{x}_b)} = \exp(f_\theta(\mathbf{x}_a, t) - f_\theta(\mathbf{x}_b, t)),$$

which may vary with time.

The negative partial log-likelihood is

$$\mathcal{L}(\theta) = - \sum_{i: \delta_i=1} \left[f_\theta(\mathbf{x}_i, t_i) - \log \left(\sum_{j \in R(t_i)} \exp(f_\theta(\mathbf{x}_j, t_i)) \right) \right].$$

The predicted survival function is

$$\hat{S}(t | \mathbf{x}) = \exp(-\hat{H}_0(t) \exp(f_\theta(\mathbf{x}, t))),$$

where $\hat{H}_0(t)$ is the estimated baseline cumulative hazard.

For concordance scoring, each survival curve is summarized by

$$r_i = -\frac{1}{K} \sum_{k=1}^K \hat{S}(t_k | \mathbf{x}_i).$$

4.4 Evaluation Metrics

Model performance was assessed using two complementary metrics:

Concordance Index (C-index)

The concordance index evaluates the model's ability to correctly rank individuals by predicted risk. For two individuals i and j with observed event times (t_i, δ_i) and (t_j, δ_j) , a pair is comparable when $t_i \neq t_j$ and at least one individual experiences an event. Let η_i and η_j denote the predicted risk scores, where larger values correspond to higher predicted hazard.

A pair is *concordant* if

$$t_i < t_j \text{ and } \eta_i > \eta_j,$$

and *discordant* if the inequality directions disagree. Ties in predicted risk or event time are handled according to standard survival-analysis conventions.

The C-index is defined as

$$C = \frac{\text{number of concordant pairs}}{\text{number of comparable pairs}} = \frac{\sum_{(i,j) \in C} \mathbf{1}[(t_i < t_j) \wedge (\eta_i > \eta_j)]}{|C|},$$

where C is the set of all comparable pairs.

For the Cox model, the linear predictor $\eta_i = \boldsymbol{\beta}^\top \mathbf{x}_i$ was negated so that higher predicted values correspond to longer survival, ensuring consistency across models. The C-index therefore measures the proportion of correctly ordered survival pairs and serves as the primary discrimination metric for comparing model performance.

Mean & Integrated Brier Score

The Brier score quantifies the calibration of predicted survival probabilities by measuring the squared error between the predicted survival probability and the observed survival status at time t . For an individual i with event time T_i , event indicator δ_i , and predicted survival probability $\hat{S}(t | \mathbf{x}_i)$, the time-dependent Brier score is

$$\text{BS}(t) = \frac{1}{n} \sum_{i=1}^n \left[Z_i(t) - \hat{S}(t | \mathbf{x}_i) \right]^2,$$

where

$$Z_i(t) = \begin{cases} 1, & T_i > t, \\ 0, & T_i \leq t \text{ and } \delta_i = 1. \end{cases}$$

Because survival data are right-censored, inverse probability of censoring weights (IPCW) are applied:

$$\text{BS}_{\text{IPCW}}(t) = \frac{1}{n} \sum_{i=1}^n w_i(t) \left[Z_i(t) - \hat{S}(t | \mathbf{x}_i) \right]^2,$$

with censoring weights

$$w_i(t) = \frac{\mathbf{1}(T_i \leq t, \delta_i = 1)}{\hat{G}(T_i)} + \frac{\mathbf{1}(T_i > t)}{\hat{G}(t)},$$

where $\hat{G}(t)$ is the Kaplan–Meier estimate of the censoring distribution.

The **mean Brier score** over a grid of evaluation times $\{t_1, \dots, t_K\}$ is

$$\text{MBS} = \frac{1}{K} \sum_{k=1}^K \text{BS}_{\text{IPCW}}(t_k).$$

The **Integrated Brier Score (IBS)** summarizes calibration over the full time horizon $[0, \tau]$:

$$\text{IBS} = \frac{1}{\tau} \int_0^{\tau} \text{BS}_{\text{IPCW}}(t) dt.$$

Lower values of both the mean Brier score and the IBS indicate better calibration and closer agreement between predicted and observed survival outcomes. For RSF and Cox-Time, the predicted survival curves $\hat{S}(t | \mathbf{x})$ were used directly, while for CoxPH, survival curves were constructed from the estimated baseline hazard and the fitted linear predictor.

5 Results

The results presented in this section evaluate model performance across three complementary dimensions: (1) the overall shape and variability of predicted survival curves, illustrated through fan charts; (2) quantitative predictive accuracy using concordance index and Brier Score metrics; (3) model calibration at a fixed time horizon; and (4) interpretability through SHAP value analysis. Together, these components provide a comprehensive assessment of how each model behaves, how well it predicts lapse risk, and which features most strongly influence those predictions.

5.1 Survival Curve Fan Charts

Fan charts provide a high-level view of how each model generates survival trajectories across the portfolio. Each line represents the predicted survival function for an individual policy, allowing us to observe the structural behavior imposed by each modeling approach.

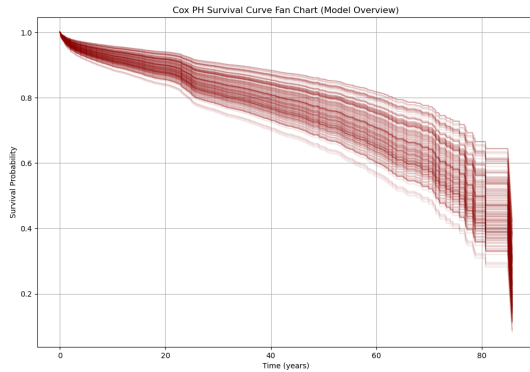


Figure 1: Fan chart of Cox-PH predicted survival curves.

5.1.1 Cox Proportional Hazards. Figure 1 displays the CoxPH fan chart. The survival curves exhibit a highly uniform shape, differing primarily in vertical displacement rather than in trajectory. This pattern reflects the proportional hazards assumption: covariate effects are constrained to be time-invariant, so all policies share

the same underlying hazard structure. As duration increases, the curves begin to diverge slightly due to reduced sample support at longer horizons, but the overall homogeneity remains a defining characteristic of the model.

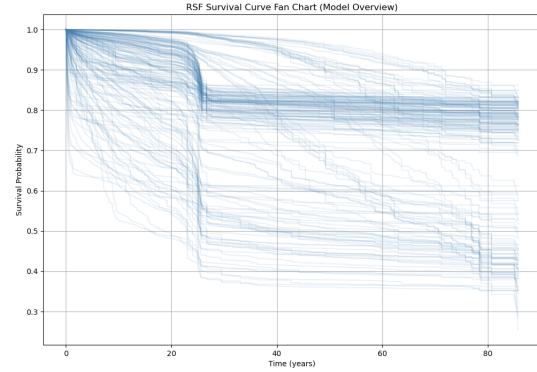


Figure 2: Fan chart of Random Survival Forest predictions.

5.1.2 Random Survival Forest. Figure 2 shows the RSF fan chart. Compared to CoxPH, RSF produces substantially greater variation in curve shapes, capturing nonlinear interactions and heterogeneous risk patterns across policies. A notable feature is a consistent decline in survival around year 25, visible across many curves. This behavior likely reflects underlying portfolio structure—such as a concentration of lapses or a policy milestone—rather than a modeling artifact. The broader spread of curves at earlier durations indicates that RSF is sensitive to early-stage behavioral differences that CoxPH cannot represent.

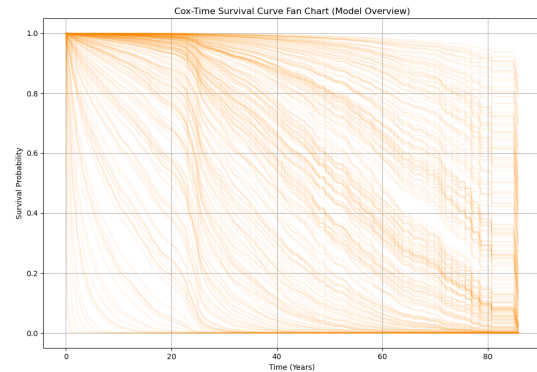


Figure 3: Fan chart of Cox-Time predicted survival curves.

5.1.3 Cox-Time. Figure 3 presents the Cox-Time fan chart. This model exhibits the greatest degree of variability, with survival curves that differ both in level and shape. Some policies show elevated early hazard, while others decline later or remain stable

Table 1: Model Performance: Concordance Index

Model	Train	Validation	Test
CoxPH	0.8358	0.8353	0.8340
RSF	0.9069	0.9015	0.8978
Cox-Time	0.9167	0.9177	0.9127

Table 2: Model Calibration: Mean and Integrated Brier Scores

Model	Mean BS	IBS
CoxPH	0.1305	0.1298
RSF	0.1039	0.1023
Cox-Time	0.0396	0.0784

throughout the horizon. Because Cox-Time models time-dependent covariate effects, it can adjust hazard dynamically as duration increases, resulting in more expressive and individualized survival trajectories. This flexibility anticipates the strong predictive performance observed in subsequent quantitative metrics.

5.2 Predictive Performance Metrics

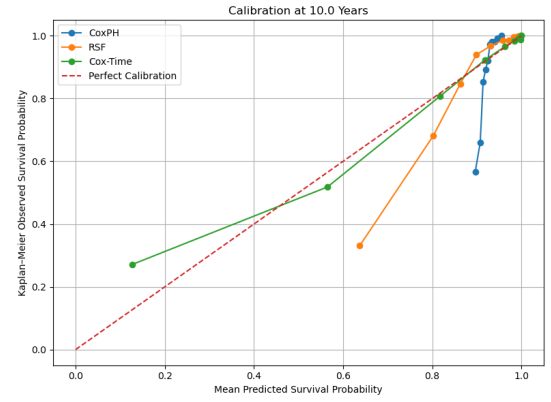
The concordance index results demonstrate clear separation among the three models. CoxPH provides stable baseline performance, but its discriminatory ability is limited by the proportional hazards structure. RSF improves substantially, reflecting its ability to capture nonlinear interactions and heterogeneous risk patterns. Cox-Time achieves the highest C-index across all data splits, indicating the strongest ability to correctly rank policies by lapse risk.

The Brier Scores reinforce this pattern. RSF reduces probability error relative to CoxPH, while Cox-Time achieves the lowest mean Brier Score and a competitive integrated Brier Score. These results indicate that Cox-Time not only ranks policies effectively but also produces more accurate survival probability estimates across the full time horizon.

5.3 Model Calibration

Figure 4 evaluates calibration at the 10-year horizon by comparing predicted survival probabilities with Kaplan–Meier estimates. A well-calibrated model aligns closely with the diagonal reference line.

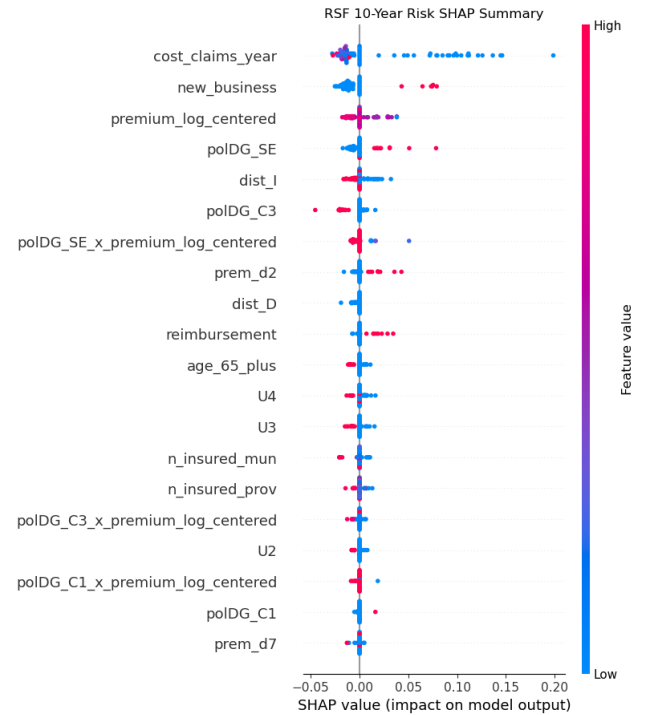
Cox-Time exhibits the strongest calibration, with predictions closely matching observed outcomes across the entire probability range. RSF performs well for low-risk policies but begins to over-estimate survival for higher-risk groups, as indicated by its curve falling below the diagonal. CoxPH shows the greatest deviation, over-predicting survival earlier and diverging from the diagonal more substantially. These calibration patterns mirror the behaviors observed in the fan charts: Cox-Time adjusts hazard more aggressively when risk increases, while RSF and CoxPH produce smoother, more conservative trajectories.

**Figure 4: Calibration plot comparing predicted and observed 10-year survival probabilities for all three models.**

5.4 Feature Importance via SHAP Values

SHAP summary plots provide insight into which features most strongly influence predicted lapse risk and the direction of their effects.

For RSF (Figure 5), `cost_claims_year` emerges as the most influential feature, with lower claim costs associated with higher predicted lapse risk. New-business status also contributes positively to lapse risk, while utilization groups generally reduce it.

**Figure 5: SHAP summary plot for the Random Survival Forest model at the 10-year horizon.**

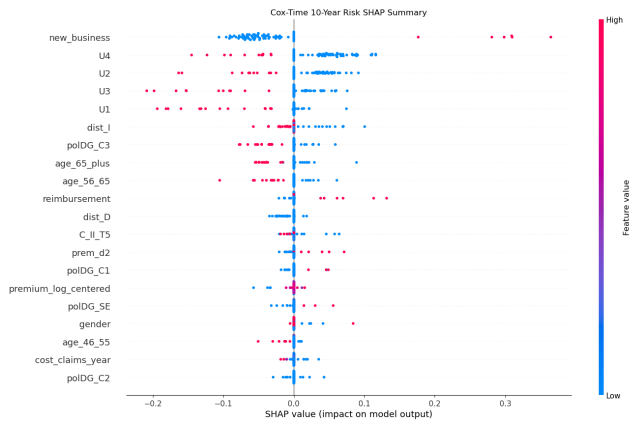


Figure 6: SHAP summary plot for the Cox-Time model.

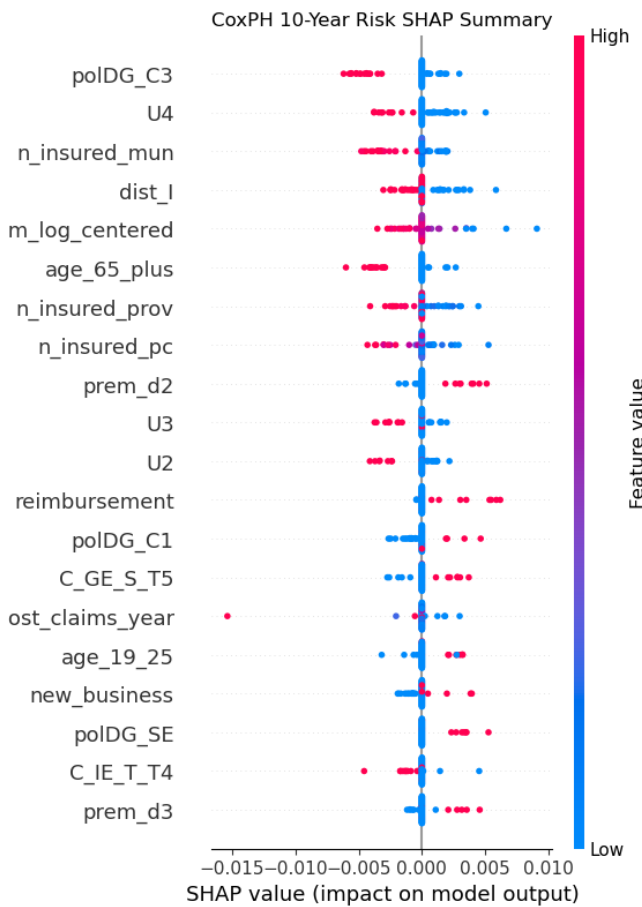


Figure 7: SHAP summary plot for the Cox Proportional Hazards model.

Cox-Time (Figure 6) exhibits the largest SHAP magnitudes, consistent with its flexible modeling capacity. New-business status strongly increases predicted lapse risk, while utilization groups

U1–U4 reduce it. Older age groups also decrease predicted lapse risk, suggesting stronger retention among older policyholders.

CoxPH (Figure 7) displays more compressed SHAP values due to its linear and time-invariant structure. The most influential features include collective policy type (polDG_C3), high utilization (U4), insured counts, distribution channel, and centered premium.

5.5 Summary of Findings

Across all evaluation dimensions—survival curve behavior, predictive accuracy, calibration, and interpretability Cox-Time demonstrates the strongest overall performance. RSF provides meaningful improvements over CoxPH and captures nonlinear structure effectively, while CoxPH remains interpretable but limited by its proportional hazards assumption. The combined evidence indicates that time-dependent hazard modeling offers substantial advantages for lapse prediction in this portfolio.

6 Discussion

The results of this study highlight clear differences in how traditional, tree-based, and neural survival models capture lapse behavior within a large health insurance portfolio. Although the empirical analysis is grounded in health insurance data, the modeling framework is fully transferable to other insurance lines. In life, disability, long-term care, or property and casualty insurance, the underlying survival models remain applicable; only the feature set and domain-specific covariates would change. This flexibility underscores the broader relevance of survival modeling for understanding policyholder behavior across the insurance industry.

6.1 Model Behavior and Structural Differences

The fan charts provide an intuitive illustration of the modeling assumptions that drive each method. CoxPH produces highly uniform survival trajectories, reflecting its proportional hazards structure. Because covariate effects are constrained to be constant over time, the model can only shift curves vertically, not alter their shape. This leads to stable but limited expressiveness, particularly when lapse behavior varies across durations.

RSF relaxes these constraints by learning nonlinear interactions and heterogeneous risk patterns. Its survival curves exhibit greater dispersion, especially at early durations, indicating sensitivity to behavioral differences that CoxPH cannot represent. The consistent decline around year 25 suggests that RSF identifies structural patterns in the portfolio, potentially related to policy anniversaries, benefit changes, or concentration of lapses.

Cox-Time extends flexibility further by allowing covariate effects to vary with time. Its survival curves differ both in level and shape, with some policies showing elevated early hazard and others declining later. This dynamic behavior aligns with the intuition that lapse risk evolves as policyholders age, accumulate claims experience, or transition between product states. The model's ability to adjust hazard as duration increases explains its superior predictive performance.

6.2 Predictive Accuracy and Calibration

The quantitative metrics reinforce the visual patterns. CoxPH performs well as a baseline, but its discriminatory ability is limited

by its structural assumptions. RSF improves substantially in both C-index and Brier Score, demonstrating the value of nonlinear modeling for lapse prediction. Cox-Time achieves the strongest performance across all metrics, indicating that time-dependent hazard modeling provides meaningful gains in both ranking ability and probability accuracy.

Calibration results further differentiate the models. Cox-Time aligns closely with observed 10-year survival probabilities, suggesting that its flexible hazard structure allows it to adjust predictions appropriately across risk levels. RSF performs well for low-risk policies but overestimates survival for higher-risk groups, consistent with its smoother hazard adjustments. CoxPH exhibits the greatest mis-calibration, particularly at higher risk levels, reflecting its inability to adapt hazard dynamically.

Importantly, calibration can be evaluated at any duration, not only at the 10-year horizon shown here. A full year-by-year calibration analysis would provide a richer understanding of how each model behaves across the entire policy lifecycle, revealing whether mis-calibration emerges early, late, or consistently over time.

6.3 Interpretability and Feature Influence

SHAP value analysis provides insight into the drivers of predicted lapse risk. RSF identifies annual claim costs and new-business status as dominant predictors, with utilization patterns and age groups also contributing meaningfully. CoxPH highlights collective policy type, utilization, and premium-related features, but the magnitude of effects is compressed due to its linear structure.

Cox-Time exhibits the largest SHAP magnitudes, reflecting its capacity to model complex, time-varying relationships. New-business status strongly increases predicted lapse risk, while utilization and older age groups reduce it. These findings align with actuarial intuition: policies with claims activity tend to remain in force, and older policyholders exhibit stronger retention.

The SHAP plots presented here evaluate feature influence at the 10-year horizon, but the same analysis can be repeated at each duration. A full time-dependent SHAP study would reveal how the importance and direction of covariate effects evolve over the life of a policy, offering a more nuanced understanding of lapse behavior and potentially uncovering duration-specific drivers of risk.

6.4 Implications for Lapse Modeling

The combined evidence suggests that Cox-Time offers the most accurate and well-calibrated predictions for this portfolio. Its ability to model time-dependent hazard is particularly valuable in lapse modeling, where risk evolves with policy duration, claims experience, and demographic changes. RSF provides a strong alternative when interpretability and robustness are prioritized, while CoxPH remains useful as a baseline or when transparency is essential.

From an operational perspective, insurers must balance predictive accuracy with interpretability and computational cost. Cox-Time delivers superior performance but requires more complex training and monitoring. RSF offers a middle ground, capturing nonlinear structure while remaining relatively stable and interpretable. CoxPH provides transparency but may underperform when lapse behavior deviates from proportional hazards.

Overall, the results demonstrate that modern survival models—particularly those incorporating time-dependent effects—can substantially improve lapse prediction and provide richer insights into policyholder behavior. Although this study focuses on health insurance, the modeling framework generalizes naturally to other insurance lines, with feature engineering tailored to the specific product context. These findings support the continued adoption of flexible survival modeling techniques in actuarial practice, especially in contexts where retention, profitability, and customer behavior evolve over time.

7 Future Work

Several extensions could further improve the modeling framework and expand the practical applicability of the results. First, additional data sources could be incorporated to enrich the covariate space. Variables such as socioeconomic trajectories, provider network characteristics, or longitudinal claim histories may capture behavioral patterns that are not fully represented in the current one-year snapshot. Incorporating multi-year claim sequences or external demographic indicators could improve the detection of early warning signals for lapse behavior.

Second, future work may explore more flexible survival models. Although Cox-Time provides time-varying effects and RSF captures nonlinearities, recent advances in deep survival modeling—including dynamic hazard networks, attention-based architectures, and transformer-based sequence models—may offer improved calibration and interpretability. These models can incorporate temporal covariates directly and may better represent complex interactions between policyholder behavior and lapse timing.

Third, model stability and fairness warrant further investigation. Because lapse behavior may vary across demographic or socioeconomic groups, future work could evaluate group-specific calibration, fairness metrics, and potential sources of bias. Techniques such as stratified modeling, group-wise hazard estimation, or adversarial debiasing could help ensure equitable performance across subpopulations.

Fourth, future work could focus on operational deployment. Integrating survival predictions into portfolio monitoring systems would require real-time scoring pipelines, threshold selection for lapse intervention strategies, and cost-benefit analyses of retention actions. Evaluating the financial impact of survival-based lapse predictions would help translate model outputs into actionable business decisions.

Finally, expanding the evaluation framework to include time-dependent AUC, calibration curves, and decision-analytic metrics could provide a more comprehensive assessment of model performance. These additional diagnostics would help identify where models perform well, where they fail, and how they might be improved in future iterations.

8 Conclusion

This study demonstrates the value of modern survival modeling techniques for predicting lapse behavior within a large health insurance portfolio. By comparing a traditional proportional hazards model, a nonlinear tree-based ensemble, and a flexible neural survival model, we show that the choice of modeling framework has

a substantial impact on predictive accuracy, calibration, and the ability to capture time-dependent risk dynamics.

CoxPH provides a stable and interpretable baseline but is limited by its assumption of time-invariant covariate effects. Random Survival Forests relax this constraint and capture meaningful nonlinear structure, improving both discrimination and probability accuracy. Cox-Time extends flexibility further by modeling time-dependent hazard, resulting in the strongest performance across all evaluation metrics and the most expressive survival trajectories. The calibration and SHAP analyses reinforce these findings, highlighting Cox-Time's ability to adjust hazard dynamically and to reveal complex, duration-specific relationships between features and lapse risk.

Although the empirical results are based on health insurance data, the modeling framework is broadly transferable across insurance lines. In life, disability, long-term care, or property and casualty insurance, the same survival models can be applied with domain-specific feature engineering. The ability to evaluate calibration and feature influence at each duration also provides a powerful foundation for understanding how lapse behavior evolves over the life of a policy, offering insurers deeper insight into retention dynamics and customer behavior.

Overall, the findings indicate that incorporating flexible, time-aware survival models—particularly Cox-Time—can substantially enhance lapse prediction and provide richer interpretability when paired with tools such as SHAP. As insurers increasingly rely on data-driven methods for retention management, pricing, and portfolio monitoring, these models offer a robust and adaptable approach for capturing the complexities of policyholder behavior over time.

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Ethics and Privacy Statement

This study uses de-identified, publicly available insurance data and does not involve any personal or sensitive information. All modeling techniques were applied in a manner consistent with responsible data use, with attention to fairness, transparency, and the avoidance of unintended harm. The methods developed here are intended for actuarial research and operational improvement, and do not pose foreseeable risks related to privacy, discrimination, or misuse.

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